

Proposed Solution to #5817 SSMJ

Solution proposed by Prakash Pant, The University of Vermont, Bardiya, Nepal.

Problem Proposed by Michel Bataille, Rouen, France.

Statement of the Problem:

Let $(F_n)_{n \geq 0}$ be the Fibonacci sequence defined by $F_0 = 0, F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for all $n \geq 2$. If m, n are integers such that $m \geq n \geq 0$, prove that

$$\sum_{j=0}^n \binom{m+1}{j} (2^{n+1} - 2^j) F_j = \sum_{j=0}^n \binom{m}{j} (2^{n+1} F_{j+2} - 2^j F_{j+3}).$$

Solution of the Problem:

$$\begin{aligned} &= \sum_{j=0}^n \binom{m+1}{j} (2^{n+1} - 2^j) F_j \\ &= \sum_{j=0}^n \left(\binom{m}{j} + \binom{m}{j-1} \right) (2^{n+1} - 2^j) F_j \\ &= \sum_{j=0}^n \binom{m}{j} 2^{n+1} F_j + \sum_{j=0}^n \binom{m}{j-1} 2^{n+1} F_j - \sum_{j=0}^n \binom{m}{j} 2^j F_j - \sum_{j=0}^n \binom{m}{j-1} 2^j F_j \\ &= \sum_{j=0}^n \binom{m}{j} 2^{n+1} F_j + \sum_{j=0}^n \binom{m}{j} 2^{n+1} F_{j+1} - \sum_{j=0}^n \binom{m}{j} 2^j F_j - \sum_{j=0}^n \binom{m}{j} 2^{j+1} F_{j+1} \end{aligned}$$

Combining first two terms and last two terms

$$= \sum_{j=0}^n \binom{m}{j} 2^{n+1} (F_j + F_{j+1}) - \sum_{j=0}^n \binom{m}{j} 2^j (F_j + 2F_{j+1})$$

Since $F_j + F_{j+1} = F_{j+2}$ and $F_j + 2F_{j+1} = F_{j+2} + F_{j+1} = F_{j+3}$, we have

$$= \sum_{j=0}^n \binom{m}{j} 2^{n+1} F_{j+2} - \sum_{j=0}^n \binom{m}{j} 2^j F_{j+3}$$

proving the initial claim.